Project Proposal

Energy-Aware Adaptive IoT Sensing Using Edge Intelligence

1. Title

Predictive Adaptive Sampling for Energy-Efficient Edge IoT: A Quantitative Trade-off Study Against Fixed-Rate and Threshold-Based Baselines

2. Problem

Conventional IoT sensor nodes sample and transmit data at a fixed rate regardless of whether the monitored condition is stable or actively changing. This wastes energy and communication bandwidth during long stable periods — a serious limitation for battery-powered deployments where node lifetime, not raw compute power, is the binding constraint. A common improvement, threshold-based adaptive sampling, only reacts once a fixed threshold has already been crossed, which means the system responds to a change after it has already begun rather than anticipating it. This project develops an IoT sensing node where the edge device dynamically adjusts its sampling rate, transmission frequency, and processing frequency based on the monitored condition, using a lightweight predictive mechanism rather than a simple reactive threshold.

3. Motivation

Battery-powered IoT nodes are deployed in large numbers across industrial, agricultural, and environmental monitoring settings, and battery replacement at scale is expensive and often impractical (e.g., buried sensors, remote field deployments, embedded structural monitors). Reducing energy consumption without sacrificing the ability to detect meaningful events directly extends the operational lifetime of these deployments and reduces both maintenance cost and electronic waste. At the same time, reducing unnecessary transmissions lowers network congestion and bandwidth cost in larger sensor deployments where many nodes share limited spectrum or gateway capacity. A system that can tell the difference between “nothing is happening, stay quiet” and “something is about to happen, pay closer attention” delivers both energy and bandwidth savings without the blind spots of a purely reactive threshold.

4. Research Gap

A substantial body of work has demonstrated that adaptive sampling can meaningfully reduce IoT energy consumption. Reported results include sensing-energy reductions of up to 60% and transmission-energy reductions of up to 93% through similarity-based redundancy elimination combined with two-stage adaptive sampling, and network traffic reductions of up to 88% through event-driven transmission strategies compared with periodic reporting. Prediction-based approaches using models such as Exponential Double Smoothing, Least-Mean-Square adaptive filters, autoregressive models, and LSTM-based edge intelligence frameworks have each shown that forecasting expected sensor values ahead of time can reduce both sampling and transmission overhead relative to fixed-rate baselines.

However, three gaps remain relevant to this project:

1. Most widely-deployed adaptive schemes are still threshold-triggered rather than genuinely predictive. The literature on prediction-based methods (EDSAS, LSTM/RL edge intelligence, autoregressive sampling) tends to target wireless sensor network research testbeds rather than low-cost, readily reproducible microcontroller platforms such as the ESP32 that student and small-deployment projects typically use.
2. Energy and accuracy results are usually reported at a single operating point, rather than as a full trade-off curve that shows how detection accuracy degrades as energy savings increase. This makes it difficult for a practitioner to choose an operating point suited to their own risk tolerance.
3. Few studies combine a genuinely predictive trigger, a real induced physical event (not only simulated or public datasets), and full multi-metric reporting (energy, bandwidth, accuracy, and latency together) on the same low-cost hardware platform.

This project addresses these gaps directly by implementing a lightweight, on-device predictive trigger, evaluating it against fixed-rate and threshold-based baselines under real induced physical events, and reporting the complete accuracy-energy-bandwidth-latency trade-off rather than a single summary number.

5. Objectives

1. Design and build an edge sensing node capable of dynamically adjusting sampling rate, transmission frequency, and on-device processing frequency in real time.
2. Implement three sensing policies — fixed-rate, threshold-based adaptive, and a proposed lightweight predictive-adaptive policy — on the same hardware platform for fair comparison.
3. Collect real experimental sensor data under both stable and deliberately induced varying conditions, with precise ground-truth event timestamps.
4. Quantify and compare energy consumption, transmitted data volume, event-detection accuracy, and detection latency across all three policies.
5. Produce a full accuracy-versus-energy trade-off curve (not a single operating point) by varying the sensitivity of the predictive trigger.

6. Methodology

Overview of approach

The system will be built around an edge node that runs three interchangeable sensing policies in turn, under identical experimental conditions, so their energy, bandwidth, accuracy, and latency can be compared fairly.

Sensing policies

- **Fixed-rate (baseline 1):** constant sampling/transmission interval, representing conventional practice.
- **Threshold-based adaptive (baseline 2):** low-rate sampling during normal readings, switching to high-rate sampling once a fixed amplitude threshold is crossed — representing the common “smart” baseline found in existing deployments.
- **Predictive adaptive (proposed):** a lightweight on-device trend/error estimator (e.g., an exponential-smoothing or simple regression-based predictor in the spirit of EDSAS/LMS-style approaches from the literature, adapted for microcontroller deployment) that estimates the likelihood of an approaching event from recent readings and pre-emptively increases sampling rate before a hard threshold would trigger.
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Experimental design

1. Validate the predictive trigger logic in software simulation using a public or self-collected reference time series before hardware deployment.
2. Run all three policies under identical induced stable/event sequences on the physical rig, repeating trials to account for variability.
3. Log ground-truth event timing precisely (operator-controlled event induction) to score detection latency and accuracy objectively.
4. Sweep the sensitivity of the predictive trigger across multiple settings to generate the accuracy-vs-energy trade-off curve.
5. Run an ablation comparing the predictive policy with and without the trend-based pre-emption component, to isolate how much benefit comes specifically from prediction versus adaptive sampling in general.

Evaluation metrics

- Energy consumed per trial and per correctly-detected event (mJ/mAh)
- Number of transmitted packets and total bytes transmitted
- Detection accuracy, precision, recall, and false-alarm rate
- Detection latency (event onset to high-rate sampling triggered)
- Projected battery lifetime under each policy for a reference battery capacity

7. Hardware

Component	Purpose
ESP32 (or ESP32-S3) development board with deep-sleep support	Main compute unit; runs all three sensing policies and adaptive logic
Accelerometer (ADXL345 or MPU6050) or gas sensor (MQ-2/MQ-135)	Primary monitored quantity — chosen for a clear, inducible stable-vs-event distinction
Current-sense resistor + INA219 breakout	Direct, ground-truth energy/current measurement during operation
Small vibration source (DC motor) or controlled gas sample	Used to induce repeatable “event” conditions on demand
18650 Li-ion battery pack and holder	Enables a realistic, hardware-based battery-life projection
Breadboard, jumper wires, USB cable	General prototyping
Raspberry Pi or laptop running a local MQTT broker (Mosquitto)	Receives transmissions and logs data for offline analysis

8. Software

- **Firmware (ESP32, C/C++ via Arduino/ESP-IDF):** implements all three sensing policies, deep-sleep/wake scheduling, sensor sampling, and the on-device predictive trigger logic.
- **Communication:** MQTT client on the ESP32 publishing readings/events to a local Mosquitto broker; message count and payload size are logged directly to measure bandwidth usage per policy.
- **Data logging and analysis:** Python (pandas, matplotlib) for processing logged sensor data, MQTT traffic logs, and INA219 current readings into the accuracy/energy/latency metrics described above.
- **Predictive model:** a lightweight statistical trend estimator (e.g., exponential smoothing or a small regression model) implemented directly in firmware; deliberately kept simple enough to run on the ESP32 without a full ML runtime, consistent with the resource constraints of the target hardware.
- **Version control and reproducibility:** firmware, analysis scripts, logged datasets, and experimental configuration files maintained in a shared repository so the experiment can be reproduced independently.

9. Expected Contribution

A quantitatively validated, lightweight predictive adaptive-sampling policy for microcontroller-class IoT nodes that measurably reduces energy consumption and transmitted data volume beyond threshold-based adaptive sampling, evaluated on real hardware against real induced physical events rather than only simulated or public datasets. The project reports the full accuracy-versus-energy trade-off curve — rather than a single operating point — directly addressing the single-point reporting and threshold-only limitations identified in the current adaptive-sampling literature, and provides a reproducible low-cost reference design (under $50 in hardware) that others can replicate.

10. Initial References

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